

SAMPLE DATA 3

INCIDENT TITLE : Production Authentication Failure After JWT Secret Rotation

INCIDENT DESCRIPTION : Authentication requests began failing after a deployment rotated the JWT signing secret. Configuration synchronization failed, causing services to validate tokens using different secrets.

INCIDENT TIMELINE :

10:00 Deployment v5.4.1 completed
10:05 Authentication failures increased
10:08 Customer login complaints received
10:12 HTTP 401 errors exceeded threshold
10:15 High priority incident declared
10:20 Investigation started
10:35 Secret synchronization issue identified
10:40 Rollback initiated
10:52 Authentication recovered

ERROR LOGS:

2026-07-22 10:05:14 ERROR JWT signature verification failed
2026-07-22 10:05:19 WARN Authentication token invalid
2026-07-22 10:06:11 ERROR Unauthorized request returned 401
2026-07-22 10:08:32 WARN User session validation failed
2026-07-22 10:12:44 ERROR Token decode exception

GIT DIFF :

diff --git a/config/security.py b/config/security.py

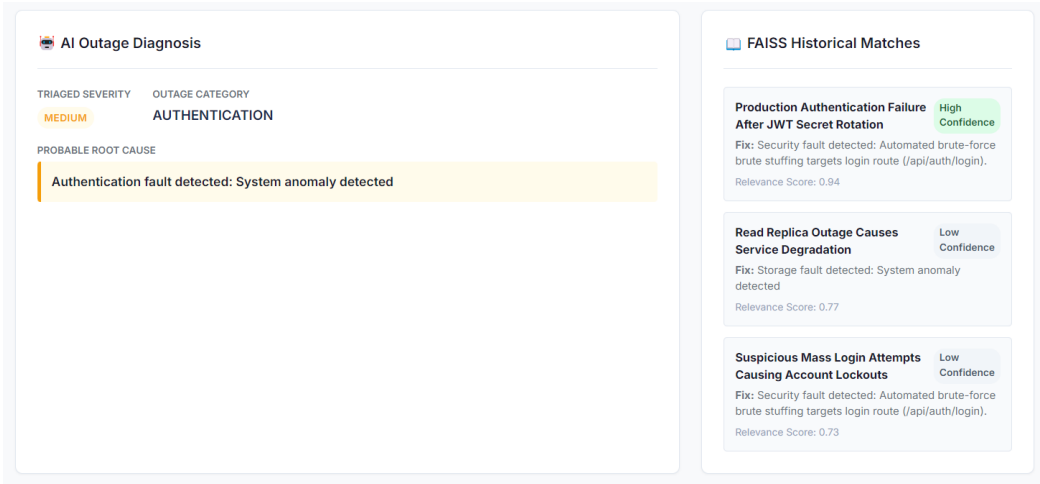
- JWT_SECRET=prod-secret-v1
+ JWT_SECRET=prod-secret-v2

- ENABLE_SECRET_SYNC=true
+ ENABLE_SECRET_SYNC=false

- TOKEN_VALIDATION_MODE=strict
+ TOKEN_VALIDATION_MODE=distributed

OUTPUTS

SEVERITY AND CATEGORY CLASSIFICATION, FAISS SIMILARITY RESULTS :



AI RESOLUTION RECOMMENDATION :

Incident Resolution Walkthrough: Production Authentication Failure After JWT Secret Rotation

Step 1: Containment (Already Completed)

The incident has been declared High Priority, and the engineering investigation has begun. The authentication failure rate has been increasing, and the first customer complaints have been received.

Step 2: Investigation and Analysis

Review the log telemetry data to identify the root cause of the authentication failures. The logs indicate JWT signature verification failures and authentication token invalid errors. Analyze the recent Git diff/code changes, which reveal that the JWT signing secret was rotated during deployment, but the configuration synchronization failed.

Step 3: Configuration Synchronization

Identify the application instances that are still using the previous JWT signing secret. Synchronize the configuration across all instances to ensure that they are using the new secret. Update the ENABLE_SECRET_SYNC configuration to true to ensure that secrets are synchronized across all instances.

Step 4: Secret Synchronization

Complete the secret synchronization process to ensure that all application instances are using the new JWT signing secret. This can be done by re-deploying the affected instances or by updating the configuration manually.

Step 5: Rollback and Verification

Roll back the changes made during deployment v5.4.1 to ensure that the authentication process is working correctly. Verify that the authentication success rate has normalized and that services are stable.

Step 6: Monitoring and Alerting

Review the monitoring and alerting configurations to ensure that similar incidents are detected and alerted in a timely manner. Adjust the baseline threshold for failed authentication attempts to prevent false positives.

Step 7: Incident Review and Post-Mortem

Conduct a thorough review of the incident, including the root cause, response, and resolution. Document the incident and identify areas for improvement to prevent similar incidents in the future.

Recommendations

Based on the incident analysis, consider implementing the following:

- Regularly review and update security configurations to prevent similar attacks.
 - Implement automated configuration synchronization to ensure that all application instances are using the same secrets and configurations.
 - Use automated tools to detect and prevent sensitive data exposure.
 - Implement a robust login rate-limiting policy and automated fail2ban blocking to slow down automated attacks.
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Action Items

- Update the `ENABLE_SECRET_SYNC` configuration to true to ensure that secrets are synchronized across all instances.
 - Review and update the `JWT_SECRET` configuration to ensure that it is correctly rotated and synchronized.
 - Implement automated configuration synchronization and monitoring to prevent similar incidents in the future.
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Diagnostic Commands

No specific technology is mentioned in the target incident data or logs, so no technology-specific diagnostic commands are provided. The resolution walkthrough focuses on general recommendations and best practices for resolving the incident.

POST INCIDENT RCA ANALYSIS :

Root Cause Analysis

Executive Summary

The incident, titled "Production Authentication Failure After JWT Secret Rotation," occurred on 2026-06-10, resulting in a significant increase in failed authentication requests across the production environment. The root cause analysis reveals that a configuration synchronization failure, following the rotation of a JWT signing secret during deployment v5.4.1, led to the incident. This report provides a detailed analysis of the incident, including its business impact, timeline, evidence collected, root cause, contributing factors, resolution, preventive actions, and lessons learned.

Business Impact

- Users Affected: Approximately 4,200 users were affected, with over 18,500 authentication failures recorded during the incident.
- Systems Affected: The authentication system, customer-facing services, and user accounts were directly affected.

- Duration: The incident lasted for approximately 52 minutes, from 10:00 to 10:52 UTC, with the resolution process completing by 11:36:17.853979+05:30 on 2026-06-10.
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Timeline

The incident timeline is as follows:

- 10:00: Deployment v5.4.1 completed.
 - 10:05: Authentication failures increased.
 - 10:08: Customer login complaints received.
 - 10:12: HTTP 401 errors exceeded threshold.
 - 10:15: High priority incident declared.
 - 10:20: Investigation started.
 - 10:35: Secret synchronization issue identified.
 - 10:40: Rollback initiated.
 - 10:52: Authentication recovered.
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Evidence Collected

Logs

Telemetry logs indicate a significant increase in failed authentication requests, with warnings for authentication token invalidation and unauthorized requests.

Example log entries include:

- 2026-07-22 10:05:14 ERROR JWT signature verification failed
- 2026-07-22 10:05:19 WARN Authentication token invalid
- 2026-07-22 10:06:11 ERROR Unauthorized request returned 401

Git Changes

Recent code changes include the rotation of the JWT signing secret from "prod-secret-v1" to "prod-secret-v2" and the disabling of secret synchronization (ENABLE_SECRET_SYNC=false).

Relevant diff entries:

- diff --git a/config/security.py b/config/security.py

Metrics

The incident resulted in over 18,500 authentication failures during the 52-minute period.

Root Cause

The root cause of the incident is a configuration synchronization failure, which occurred after the rotation of the JWT signing secret during deployment v5.4.1. This failure caused several application instances to continue using the previous secret, resulting in authentication tokens generated by one service instance being invalid for other instances.

Contributing Factors

Contributing factors to the incident include:

- The recent code changes that rotated the JWT signing secret without ensuring proper configuration synchronization across all application instances.
 - The disabling of secret synchronization (`ENABLE_SECRET_SYNC=false`), which removed a critical control that could have prevented or mitigated the incident.
 - Historical precedents, such as the "Production Authentication Failure After JWT Secret Rotation" incident, which highlighted the importance of proper configuration management and security controls.
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Resolution

The incident was resolved by:

- Initiating a rollback to the previous configuration.
 - Completing the secret synchronization across all application instances.
 - Verifying the authentication success rate and service stability.
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Preventive Actions

To prevent similar incidents in the future, the following actions will be taken:

- Implementing automated configuration synchronization and validation for all application instances.
 - Enabling secret synchronization (`ENABLE_SECRET_SYNC=true`) to ensure that all instances use the same secrets.
 - Conducting regular security audits and penetration testing to identify vulnerabilities and weaknesses in the system.
 - Developing and implementing a comprehensive incident response plan to quickly respond to and resolve future incidents.
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Lessons Learned

The incident highlights the importance of proper configuration management, secret synchronization, and security controls in preventing and mitigating authentication failures. The lessons learned from this incident will be applied to improve the overall security and reliability of the system, including:

- Ensuring that all configuration changes are thoroughly tested and validated before deployment.
- Implementing automated monitoring and alerting systems to quickly detect and respond to authentication failures.
- Conducting regular security training and awareness programs for developers and engineers to emphasize the importance of security and configuration management.